



Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.

UNIT 9 • STANDARD 6.AT.11

Explore Relationships Between Two Variables

Lesson 9-1 • Teacher Copy

Name: _____

Date: _____

Class: _____



TODAY'S OBJECTIVES

Content Objective: I can recognize when two quantities change together, and identify which one is the independent variable and which is the dependent variable.

Language Objective: I can explain which quantity depends on the other using the words independent variable, dependent variable, and depends on.



See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.



Fill in each blank as we go. Use the Word Bank to help you.



WORD BANK — FILL EACH BLANK WITH THE BEST WORD

independent variable

dependent variable

depends on

quantity

relationship



Tap any word to see its meaning and a picture.

1

The quantity that is subject to choice is the

2

The quantity that changes in response to the other is the

3

Say it in order: the responding quantity

the chosen one.

4

Something that can be counted or measured is a

5

How two quantities change together is a

Write About the Math

Which two quantities in this shop are related, so that one depends on the other — and which quantity does NOT depend on the number of pizzas at all?

¿Qué dos cantidades de esta pizzería están relacionadas, de modo que una depende de la otra — y qué cantidad no depende para nada del número de pizzas?

Say your answer to a partner first. Then write it, using at least two of these words:

☐ **independent variable**
variable independiente

☐ **dependent variable**
variable dependiente

☐ **depends on**
depende de

☐ **quantity**
cantidad

☐ **relationship**
relación

Model: *Akela chose where to place each tray, which set the hours of sunlight; the plant height was measured afterward, in response to the sunlight.* — A strong answer names sunlight as chosen (independent) and height as responding (dependent), and can say the sentence 'height depends on sunlight.' A weak answer just says 'height is bigger' without naming which quantity was chosen.

Start

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

- **In this scenario, the ____ is the independent variable. The ____ and the ____ both depend on it. The number of ____ stays the same no matter how many pizzas are made.**
- **My answer is ____ because ____.**

Mi respuesta es ____ porque ____.

Build

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- **My claim is ____.**

Mi afirmación es ____.

- **My evidence is** ____, **so** ____.
- Mi evidencia es* ____, *entonces* ____.

Explain

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

☐ I answered the question.

☐ I used math evidence: numbers, an equation, or a model.

☐ I used at least two math words.

Writing Guide — Teacher Copy

- The answer to the math question is correct.
- The evidence is real lesson mathematics (numbers, equation, or model), not restated claim.
- The support level changed the language scaffolding, never the mathematical expectation.

Answer Key & Teacher Guide

1. **Notes 1:** independent variable
2. **Notes 2:** dependent variable
3. **Notes 3:** depends on
4. **Notes 4:** quantity
5. **Notes 5:** relationship
6. **Watch for this mistake:** *Naming the variables backward — calling the responding quantity 'independent' because it is the one the question asks about.*
7. **Try It 1:** A. A student's height and their house number — *A house number is assigned by an address, not by how tall someone is — the two are unrelated.*
8. **Try It 2:** A. Pages printed depends on minutes running — *You choose how long to let it run; the number of pages is the result of that choice.*